Intro to cyber threat Intel

Threat Intelligence provides the context that helps an analyst decide which of two hundred alerts represents genuine danger.

CTI seeks to answer three essential questions:

1) Who or what, is on the other end of this alert indicator?
2) What was their Behaviour in the past?
3) How does my organisation respond, and what should I do about it right now?

Labels:

- Indicator of Compromise (IOC): Evidence of a Breach
- Indicator of Attack (IOA): A malicious action
- Tactics, Techniques and Procedures (TTP)
  ↳ An adversary's detailed methodologies expressed in MITRE ATTACK IDs and descriptions

Tactics: attacker's goal, such as execution, persistence, many

Techiques / Procedures: the exact methods they use to achieve that goal

Threat Intelligence Classifications

Strategic intel: High level intelligence that looks into the organisation's threat landscape and maps out the risk areas based on trends, patterns and emerging threats that may impact business decisions.

- Example: annual ransomware trends report predicting a shift to data-wiping exortion in healthcare

Tactic Intel: Assessments of adversaries' Behaviours through analysis of tactics, techniques and procedures TTPs.

- Basically which means focuses on how attackers operate, not just who they are or what malware they use.

Operational Intel: Campaign-specific details about the motives and intent to perform an attack. This is useful for understanding the critical assets available in the organisation that may be targeted.

Technical Intel: Atomic indicators and artefacts such as IPs and hashes related to an attack

Six-phase ~~intelg~~ intelligence lifecycle :

1) Planning and Direction

2) Collection

3) Processing

4) Analysis

5) Dissemination

6) Feedback

File and Hash Threat Intel

File hashes uniquely identify files and malware, regardless of name changes

Windows:

Command prompt: certutil -hashfile file.exe SHA256

Powershell: Get-Filehash -Algorithm SHA-256 file.exe

Linux: sha256sum file.exe

MalwareBazaar is an all in one malware collection and database

- Security analysts can upload their malware samples for analysis and build the intelligence database

- Hunting for malware samples is possible through:

  - We can find files classified by their malware families
  - YARA rule integration: submissions include detection rule
  - Tags which belongs to a threat actor group
  - Samples are available for download and analysis

Security analysts will use sandboxes to do 3 things:

- Confirm execution
- Extract runtime Indicators of Compromise
- Map to Att&ck

Sandbox limitations:

1) Sandbox evasion techniques
2) Limited Execution time & Coverage
3) Encrypted & Obfuscated traffic
4) Fileless & Living-off-the-Land Malware

Many sand Boxes cannot decrypt SSL/TLS traffic, leading to blindspot. This may result in HTTPS C2 Traffic appearing with no payload visibility or the malware utilising DNS tunnelling to hide data in DNS queries

LoL Threats never touch disk